

L03: System Boundaries and Model Scope

Simulation Without a Black Box

Outline

Learning Objectives

Entity–Resource–Event–State Framework

Boundary Choice

Endogenous vs. Exogenous Variables

Worked Example: Primary-Care Clinic

Dangers of Over- and Under-Scoping

Activity: Port Terminal

Key Takeaways

Learning Objectives

By the end of this lecture you will be able to:

- Apply the entity–resource–event–state framework to a new system.
- Choose a system boundary (narrow, medium, or wide) and defend the trade-offs.
- Distinguish endogenous from exogenous variables and explain why the distinction matters.
- Identify the risks of over-scoping and under-scoping a model.

The Four-Element Framework

Entities

Objects that flow through the system and whose experiences we track.

Examples: patients, ships, phone calls, jobs.

Resources

Capacitated elements that entities compete for.

Examples: physicians, cranes, agents, machines.

A conceptual model is complete when you can specify all four elements unambiguously.

Events

Instantaneous occurrences that change the system state.

Examples: arrival, service-begin, service-end, breakdown.

State variables

The minimal set of information needed to resume the simulation at any point.

Examples: queue length, server status, clock time.

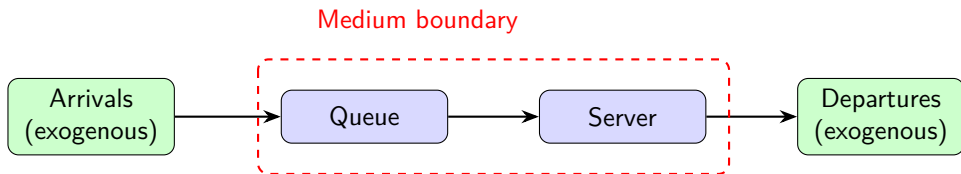
Three Levels of Model Boundary

Scope	Includes	Advantage	Risk
Narrow	Core queue only	Fast to build, easy to validate	Misses upstream/downstream effects
Medium	Core + immediate upstream/downstream	Balanced; usually appropriate	Requires more data
Wide	Full supply chain or network	Captures system-level effects	Expensive; hard to validate

Rule of thumb

Start narrow. Expand the boundary only when a *specific decision* requires it.

Boundary as a TikZ Diagram



Arrivals and departures are **exogenous**: they are inputs to, not outputs of, the model. Queue length and server utilization are **endogenous**: the model computes them.

Endogenous vs. Exogenous Variables

Exogenous (inputs, driven from outside)

- Arrival rate $\lambda(t)$
- Service time distributions (from data)
- Resource capacity (given by management)
- Scheduled shift patterns

Endogenous (computed by the model)

- Queue length $L_q(t)$
- Utilization ρ
- Mean waiting time W_q
- Throughput rate

Why does this matter?

- Exogenous variables need *data* or *assumptions* — a data-collection plan.
- Endogenous variables are the *performance measures* — what the model produces.
- Confusing the two leads to circular logic or missing inputs.

Worked Example: Primary-Care Clinic

Decision: How many exam rooms are needed to keep wait times under 20 minutes?

Inside the boundary

- Check-in desk (1 clerk)
- Waiting room (unlimited seats)
- Exam rooms (resource to size)
- Physicians (3 on duty)
- Nurses (triage and vitals)

Outside the boundary

- Pharmacy (post-visit; separate system)
- Lab tests (long-delay subprocess; exogenous)
- Scheduling system (sets $\lambda(t)$; input only)
- Billing (no effect on wait time)
- Parking lot

Scope justification

The decision is about in-clinic flow. Pharmacy and labs are downstream; they affect patient satisfaction but not the wait metric being optimized.

Dangers of Over-Scoping and Under-Scoping

Under-scoping

- Ignores interactions that drive congestion.
- Example: omitting the check-in clerk when arrivals bunch.
- Result: model predicts zero wait at check-in; reality does not.
- The model answers a *different* question than the one asked.

Over-scoping

- Includes the pharmacy, lab, billing, parking.
- Requires data on lab turnaround, pharmacist staffing, etc.
- Build time doubles; most data are unavailable.
- Stakeholders lose confidence in a model they cannot understand.
- The output is noisy — signal buried in irrelevant complexity.

Both failure modes are common in practice. The cure is a clear problem statement before any modeling begins.

Activity: Draw the Boundary for a Port Terminal

Setting: A container terminal receives ships, unloads containers via cranes, moves containers by yard trucks to a storage yard, then loads outbound containers to trains.

Decision: How many cranes are needed to meet a ship-departure target of 18 hours?

Your tasks:

1. List at least 5 entities, 4 resources, and 6 events.
2. Draw a boundary diagram showing what is inside and what is exogenous.
3. Label each variable as endogenous or exogenous.
4. Justify one element you chose to *exclude* from the narrow boundary.

Compare your boundary diagram with a partner. Resolve any differences.

Key Takeaways

- Every simulation model has a boundary. Choosing it explicitly is part of modeling.
- The entity-resource-event-state framework structures conceptual thinking before coding.
- Endogenous variables are outputs; exogenous variables are inputs requiring data.
- Under-scoping misses interactions; over-scoping inflates cost and erodes trust.
- Boundary choice should be driven by the decision question, not convenience.

Next lecture: Entities, resources, events, activities, and state variables in detail (L04).